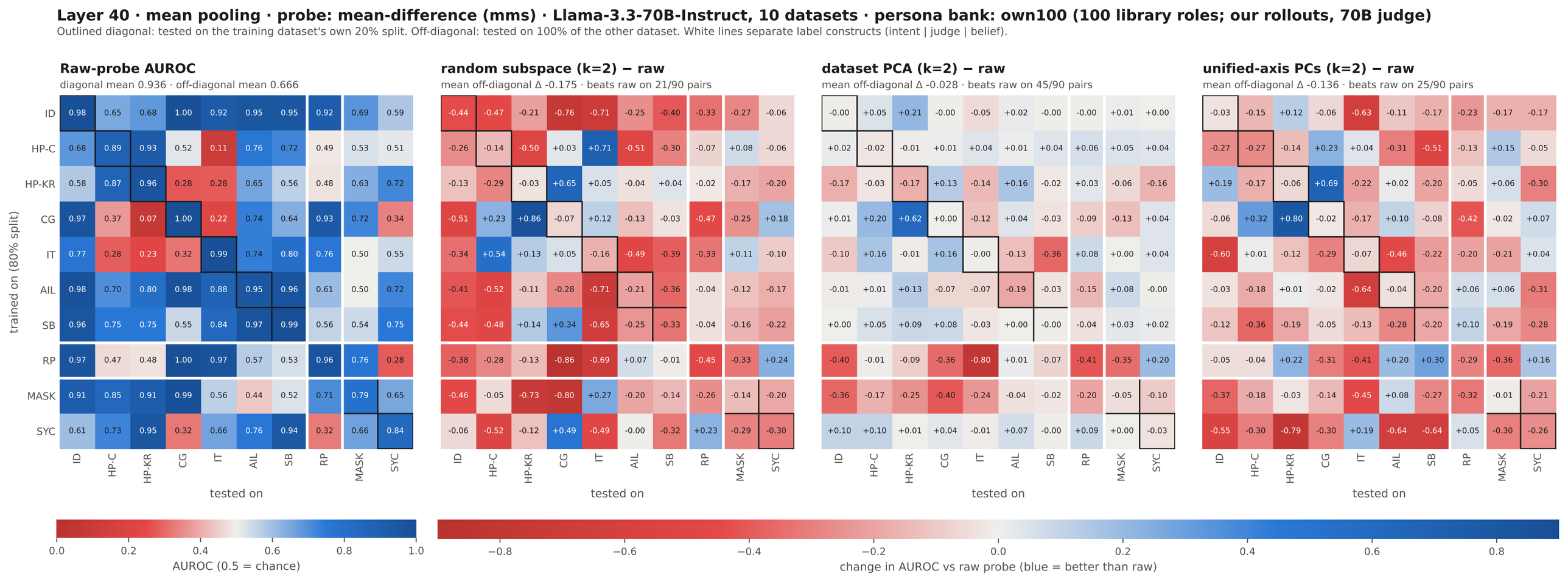

Layer 40 · mean pooling · probe: mean-difference (mms) · Llama-3.3-70B-Instruct, 10 datasets · persona bank: own100 (100 library roles; our rollouts, 70B judge)
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.936 · off-diagonal mean 0.666
random subspace (k=2) − raw
mean off-diagonal Δ -0.175 · beats raw on 21/90 pairs
dataset PCA (k=2) − raw
mean off-diagonal Δ -0.028 · beats raw on 45/90 pairs
unified-axis PCs (k=2) − raw
mean off-diagonal Δ -0.136 · beats raw on 25/90 pairs
trained on (80% split)
ID HP-C HP-KR CG IT AIL SB RP MASK SYC
tested on
AUROC (0.5 = chance)
change in AUROC vs raw probe (blue = better than raw)